

1) A company claims that 70% of its customers are satisfied with their service. A disgruntled customer believes this is an overstatement, and wants to conduct a survey to test this claim. A random sample of 200 customers is taken, and it is found that 130 of them are satisfied. Are the results significant at the 5% significance level?

<u>Communicate</u>	<u>Conditions</u>
Hypotheses:	Random:
Proportion in context:	Independent:
Significance Level:	Normal:
<u>Calculate</u>	<u>Conclude</u>
Inference Method:	
Sample proportion:	
z-score and p-value calculation:	

2) Determine if the following statements are true (T) or false (F)

- ___ a) It is helpful to examine your data before deciding whether to use a one-sided or two-sided hypothesis test.
- ___ b) If the p-value is 0.05, the probability that the null hypothesis is correct is 0.05.
- ___ c) A two-sided hypothesis test is used when we are unsure about the direction of the difference from the null hypothesis
- ___ d) The larger the p-value, the more evidence there is against the null hypothesis.
- ___ e) If the p-value is small enough, we can conclude that the alternative hypothesis is true.

3) A technology company claims that 25% of its users upgrade to the premium version of their software after the free trial. A watchdog organization wants to investigate this claim to see if they find evidence that it is not true. In a random sample of 400 users who tried the free trial, 120 upgraded to the premium version. Is this evidence that the company's claim is incorrect? Use a 5% level of significance.

4) For a hypothesis test of $H_0: p = p_0$ against the alternative $H_a: p < p_0$ the z-test statistic is found to be -1.96 . This finding is

- (A) significant at the 0.05 level but not at the 0.01 level.
- (B) significant at the 0.01 level but not at the 0.05 level.
- (C) significant at both the 0.01 and the 0.05 levels.
- (D) significant at neither the 0.01 nor the 0.05 levels.
- (E) not large enough to be considered significant.